CS 549 Distributed Systems and Cloud Computing

Stevens Institute of Technology
Summer 2021

Instructor: Dominic Duggan
Course Schedule: Monday-Sunday
Contact Info: Canvas email preferred; dduggan@stevens.edu
Virtual Office Hours: W 5:30 - 6:30 PM, F 1:00 - 2:00 PM (Zoom: https://stevens.zoom.us/j/97641369501)
Virtual session URL: Go to the “Zoom” tab on the left-hand navigation bar to access live virtual sessions for this course.
Prerequisite(s): You are expected to have a certain amount of programming maturity, since you will be developing distributed programs during the course of the term. You should know Java since that will be the language for the assignments. If you know C#, you can learn Java very quickly.
Corequisite(s): N.A.
Cross-listed with: N.A.
Credits: 3

COURSE DESCRIPTION

The objective of this course is to give students a basic grounding in designing and implementing distributed and cloud systems. Developers of cloud services question how those services should be implemented. What are global consensus and Paxos, and what are their applications in building cloud systems? What are the advantages and disadvantages of using distributed NoSQL stores such as Cassandra instead of relational stores such as MySQL? What are strong and weak consistency, what is the "CAP Theorem," and what is its implication for building highly available services? What are the roles of REST, Websockets, and stream processing in cloud applications? This

course will combine hands-on experience in developing cloud services with a firm grounding in the tools and principles of building distributed and cloud applications, including advanced architectures such as peer-to-peer, publish-subscribe, and streaming. Besides cloud services, we will also be looking at cloud support for batch processing, such as the Hadoop and Pig frameworks, and their use with NoSQL data stores such as Cassandra.

STUDENT LEARNING OUTCOMES

After successful completion of this course, students will be able to…

- Outcome 1: Services - Develop distributed and Web services, using programming paradigms such as client-server, streaming, peer-to-peer, event-driven, and REST.
- Outcome 2: Agreement - Explain protocols for achieving agreement in distributed systems, such as Two Phase Commit (2PC) and Paxos, and explain their application in scenarios such as transactional e-commerce and cloud infrastructure.
- Outcome 3: Availability - Explain methods for building highly available services, and explain the trade-off between weak and strong consistency as given in the CAP theorem. Explain the significance for cloud systems such as Cassandra and Chubby.
- Outcome 4: Cloud - Develop cloud applications using frameworks such as Cassandra, Hadoop, Pig, and Hive.
- Outcome 5: Storage - Describe implementation methods and protocols for distributed storage systems: distributed file systems, distributed hash tables, cloud storage, and NoSQL databases. Describe examples such as NFS, Dropbox, HFS, Dynamo, and Cassandra.

COURSE FORMAT AND STRUCTURE

This course is fully online. To access the course, please visit stevens.edu/canvas . For more information about course access or support, contact the Technology Resource and Assistance Center (TRAC) by calling 201-216-5500.

Class Format

1. **Classroom lectures:** You are expected to attend 60-minute meeting sessions each week, where class material and assignments will be discussed. Class will be online and at the time assigned by the registrar. It will consist of lectures, discussion and group breakout sessions. Part of your grade will depend on your participation in these sessions.
2. **Lecture slides and videos:** I will be making slides and supplementary lecture videos available each week, via Canvas. It is your responsibility to review these materials and take the exams (see below). It is important that you keep up with this material.
3. **Reading:** There will sometimes be reading associated with each topic. It is highly recommended that you do the reading. You should view the lectures as intended to draw out what is important in the reading and explain the key points of understanding. By doing the reading, you will get much better depth of understanding in the material than can be made available in the slides alone. Readings will be from the texts and from other on-line materials as the term progresses.
4. **On-line discussion:** There will be Piazza discussion groups where people can discuss the assignments. These discussions go direct to your mailbox, so they may be more useful for the "urgent" questions people may be asking as an assignment deadline nears.
5. **Assignments:** There will be approximately biweekly programming assignments. I will provide "starter" projects, that you complete. They will be due at midnight on Sunday at the end of the week in which their due date falls, via Canvas.
6. **Exams:** There will weekly "mini-exams" (more extensive than a quiz, but shorter than a full exam). They will be due on Monday the week after the material is covered in on-line lectures. There will be an online final exam that reviews the material covered by these mini-exams.
7. **On-line discussion:** There will be Piazza discussion groups where people can discuss the assignments or any part of the class material. These discussions go direct to your mailbox, so they may be more useful for the "urgent" questions people may be asking as an assignment deadline nears.

Instructor's Online Hours

I will have office hours via Canvas an hour before each weekly meeting session. I will be available via **Canvas email** and will respond as soon as I am available (generally within 24-48) hours. For the online discussions, I will check in at least 3 times per week. Often I will not need to respond to questions, as students often can answer each others' questions before I have a chance to respond. For urgent matters, feel free to email me;

Canvas email is much preferred as the Canvas mailbox is not as cluttered as regular email (and gets forwarded to my regular email anyway).

Virtual Office Hours

Virtual Office Hours are a weekly synchronous session (through Zoom) to give you the opportunity to ask questions related to course material and/or assignments.

Online Etiquette Guidelines

Your instructor and fellow students wish to foster a safe online learning environment. All opinions and experiences, no matter how different or controversial they may be perceived, must be respected in the tolerant spirit of academic discourse. You are encouraged to comment, question, or critique an idea but you are not to attack an individual. Please consider that sarcasm and humor can be misconstrued in online interactions and generate unintended disruptions. Working as a community of learners, we can build a polite and respectful course ambience. Please read the Netiquette rules for this course:

- Do not use offensive language. Present ideas appropriately.
- Be cautious in using Internet language. For example, do not capitalize all letters since this suggests shouting.
- Avoid using vernacular and/or slang language. This could possibly lead to misinterpretation.
- Keep an “open mind” and be willing to express even your minority opinion.
- Do not hesitate to ask for feedback.
- There are no “stupid” questions. If you have a question, other students probably have the same question. The only way the instructor will learn of it, is if someone asks it.

TENTATIVE COURSE SCHEDULE

Week or Module	Topic(s)	Readings	Assignment
Week/Module 01	Introduction to Distributed Systems and Cloud Computing	Required: Articles and lecture slides provided in online classroom	Quiz #1 due May 30 (Note: all assignments are due at 11:59 PM Eastern Time)

Week/Module 02	Client-Server Computing	Required:Articles and lecture slides provided in online classroom	Quiz #2 due June 6 Assignment #1 due June 6
Week/Module 03	Distributed File Systems	Required: Articles and lecture slides provided in online classroom	Quiz #3 due June 13
Week/Module 04	Web Services and Failure Models	Required: Articles and lecture slides provided in online classroom	Quiz #4 due June 20 Assignment #2 due June 20 Quiz #5 due June 20
Week/Module 05	Time and Ordering of Events	Required: Articles and lecture slides provided in online classroom	Quiz #6 due June 27
Week/Module 06	Asynchronous Communication	Required: Articles and lecture slides provided in online classroom	Assignment #3 due July 4
Week/Module 07	Transactions	Required: Articles and lecture slides provided in online classroom	Quiz #7 due July 11
Week/Module 08	Distributed Transactions	Required: Articles and lecture slides provided in online classroom	Quiz #8 due July 18 Assignment #4 due July 18
Week/Module 09	MapReduce; High Availability	Required: Articles and lecture slides provided in online classroom	Quiz #9 due July 25 Quiz #10 due July 25

Week/Module 10	Cloud Computing: Hadoop, Pig	Required: Articles and lecture slides provided in online classroom	Assignment #5 due Aug. 1
Week/Module 011	Consensus Protocols: Paxos, PBFT	Required: Articles and lecture slides provided in online classroom	Quiz #11 due Aug. 8
Week/Module 12	Blockchain Protocols	Required: Articles and lecture slides provided in online classroom	Quiz #12 due Aug. 15
Week/Module 13	Eventual Consistency and NoSQL	Required: Articles and lecture slides provided in online classroom	Final Exam due Aug. 22

COURSE MATERIALS

Note: The course materials for CS-549 are articles that are free and accessible via the Stevens library.

[BASE] "BASE: An ACID Alternative." Dan Pritchett, ACM Queue 6(3), 2008. Link.

[BT] F. Chang et al. BigTable: A Distributed Storage System for Structured Data. ACM Transactions on Computer Systems, 2008. https://doi.org/10.1145/1365815.1365816. Link.

[CALM] J. Hellerstein and P. Alvaro. Keeping CALM: When Distributed Consistency is Easy. Communications of the ACM 63(9), September 2020. https://doi.org/10.1145/3369736. Link.

[Cass] A. Lakshman, P. Malick. Cassandra: A Decentralized Structured Storage System. ACM SIGOPS Operating Systems Review, April 2010. https://doi.org/10.1145/1773912.1773922. Link.

[CL85] Chandy, K. M., & Lamport, L. (1985). Distributed snapshots: Determining global states of distributed systems. *ACM Transactions on Computer Systems 3*(1), 63–75. Link.

[Dyn] G. DeCandia et al. Dynamo: Amazon's Highly Available Key-value Store. ACM SIGOPS Operating Systems Rev., 41(6), 2007, pp. 205–220. https://doi.org/10.1145/1323293.1294281. Link.

[E2E] Saltzer, J. H., Reed, D. P., & Clark, D. D. (1984). End-to-end arguments in system design. *ACM Transactions on Computer Systems, 2*(4), 277–288. Link.

[F09] Patterson, D. A., Katz, R. H., Joseph, A. D., Fox, A., Stoica, I., Lee, G., Zaharia, M. … Rabkin, A. (2009). *Above the clouds: A Berkeley view of Cloud Computing.* (UC Berkeley Tech Report). Berkeley, CA: University of California, Berkeley. Link.

[GFS] Ghemawat, S., Gobioff, H., & Shun-Tak Leung, S-T. (2003). The Google file system. Paper presented at the ACM Symposium on Operating Systems Principles. Link.

[GL06] Gray, J., & Lamport, L. (2006). Consensus on transaction commit. *ACM Transactions on Database Systems*. Link.

[L78] Lamport, L. (1978). Time, clocks and the ordering of events in a distributed system. *Communications of the ACM 21*(7), 558–565. Link.

[MR] Dean, J., & Ghemawat, S. (2008). MapReduce: Simplified data processing on large clusters. *Communications of the ACM, 51*(1), 107–113. Link.

[NFSv4] Pawlowski, B., Shepler, S., Noveck, D., Robinson, D., & Thurlow, R. (2000). The NFS Version 4 Protocol. Paper presented at the Second International System Administration and Networking Conference (SANE). Link.

[Paxos] L. Lamport, Paxos Made Simple. ACM SIGACT News (Distributed Computing Column) 32, 4 (Whole Number 121, December 2001) 51-58. Link.

[PBFT] Liskov B. From Viewstamped Replication to Byzantine Fault Tolerance. In: Charron-Bost B., Pedone F., Schiper A. (eds) Replication. Lecture Notes in Computer Science, vol 5959. Springer, Berlin, Heidelberg, 2010. Link.

[Pig] Olston, C., Reed, B., Srivastava, U., Kumar, R., & Tomkins, A. (2008, June). Pig Latin: A not-so-foreign language for data processing. *ACM SIGMOD,* 1099–1110. Link.

[RGO08] Rotem-Gal-Oz, A. (2008). Fallacies of distributed computing explained. *Doctor Dobbs Journal.* Link.

[Session] D. B. Terry, A. J. Demers, K. Petersen, M. J. Spreitzer, M. M. Theimer and B. B. Welch, "Session guarantees for weakly consistent replicated data," Proceedings of 3rd International Conference on Parallel and Distributed Information Systems, Austin, TX, USA, 1994, pp. 140-149, doi: 10.1109/PDIS.1994.331722. Link.

[Thund] R. Pass and E. Shi. Thunderella: Blockchains with Optimistic Instant Confirmation. EUROCRYPT, 2018. Link.

[W94] Waldo, Wyant, G., Wollrath, A., & Kendall, S. (1994). *A note on distributed computing* (SMLI TR-94-29). Mountainview, CA: Sun Microsystems. Link.

TECHNOLOGY REQUIREMENTS

All assignments will be done in the Java programming language. Assignments will involve the use of tools such as:

- Eclipse IDE and Maven.
- Sockets and Java RMI (standard APIs in Java).
- Jersey framework for Web services.
- Tyrus framework for Websockets.

In addition, there will be an assignment involving the following:

- Hadoop (programming environments for cloud data processing).

GRADING PROCEDURES

Grades will be based on:

Assignment (Quiz, Discussion, etc.)	Percentage of Final Grade
Quizzes	40
Assignments	40
Participation	10
Final Exam	10

TOTAL	100

Late Policy

Assignments may be submitted after the due date, but up until the cutoff date, with a penalty of -5%. Assignments may be submitted up until the second cutoff date (usually two weeks after the original due date), with an additional penalty of -20%. Assignments may be submitted up until the third cutoff date (usually three weeks after the original due date), with an additional penalty of -25%. There will be no extensions past the third cutoff date. If the cutoff date is the same as the due date, no late extensions are allowed. Please note that an assignment with a penalty of -50% still carries a much higher grade than no assignment at all. *Please also note that resubmission of assignments is not allowed. We do not have the resources to regrade assignments, so please be sure to submit the final version when it is ready.*

All exams and quizzes ("mini-exams") must be submitted by the time and date posted. There will be no extensions and no late submissions allowed. However, assuming that N mini-exams are administered during the semester, your mini-exam grade will be based on the best N-1 of your mini-exam scores. Therefore you maximize your possible grade by taking all mini-exams, but you do not suffer any penalty if you miss a single mini-exam. It is your responsibility not to miss more than one mini-exam.

Academic Integrity

Undergraduate Honor System

Enrollment into the undergraduate class of Stevens Institute of Technology signifies a student's commitment to the Honor System. Accordingly, the provisions of the Stevens Honor System apply to all undergraduate students in coursework and Honor Board proceedings. It is the responsibility of each student to become acquainted with and to uphold the ideals set forth in the Honor System Constitution. More information about the Honor System including the constitution, bylaws, investigative procedures, and the penalty matrix can be found online at http://web.stevens.edu/honor/

The following pledge shall be written in full and signed by every student on all submitted work (including, but not limited to, homework, projects, lab reports, code, quizzes and exams) that is assigned by the course instructor. No work shall be graded unless the pledge is written in full and signed.

"I pledge my honor that I have abided by the Stevens Honor System."

Reporting Honor System Violations

Students who believe a violation of the Honor System has been committed should report it within ten business days of the suspected violation. Students have the option to remain anonymous and can report violations online at www.stevens.edu/honor.

Graduate Student Code of Academic Integrity

All Stevens graduate students promise to be fully truthful and avoid dishonesty, fraud, misrepresentation, and deceit of any type in relation to their academic work. A student's submission of work for academic credit indicates that the work is the student's own. All outside assistance must be acknowledged. Any student who violates this code or who knowingly assists another student in violating this code shall be subject to discipline.

All graduate students are bound to the Graduate Student Code of Academic Integrity by enrollment in graduate coursework at Stevens. It is the responsibility of each graduate student to understand and adhere to the Graduate Student Code of Academic Integrity. More information including types of violations, the process for handling perceived violations, and types of sanctions can be found at www.stevens.edu/provost/graduate-academics.

Special Provisions for Undergraduate Students in 500-level Courses

The general provisions of the Stevens Honor System do not apply fully to graduate courses, 500 level or otherwise. Any student who wishes to report an undergraduate for a violation in a 500-level course shall submit the report to the Honor Board following the protocol for undergraduate courses, and an investigation will be conducted following the same process for an appeal on false accusation described in Section 8.04 of the Bylaws of the Honor System. Any student who wishes to report a graduate student may submit the report to the Dean of Graduate Academics or to the Honor Board, who will refer the report to the Dean. The Honor Board Chairman will give the Dean of Graduate Academics weekly updates on the progress of any casework relating to 500-level courses. For more information about the scope, penalties, and procedures pertaining to undergraduate students in 500-level courses, see Section 9 of the Bylaws of the Honor System document, located on the Honor Board website.

EXAM CONDITIONS

The following procedures apply to quizzes and exams for this course. As the instructor, I reserve the right to modify any conditions set forth below by printing revised Exam Conditions on the quiz or exam.

1. Students may use the following materials during quizzes and/or exams. Any materials that are not mentioned in the list below are not permitted.

Material	Permitted?	
	Yes	No
Handwritten Notes Conditions: i.e., size of note sheet	Yes	
Typed Notes Conditions: i.e., size of note sheet	Yes	
Textbooks Conditions: i.e., specific books	Yes	
Readings Conditions: i.e., specific documents	Yes	
Videos provided by the instructor	Yes	

2. Students are/are not allowed to work with or talk to other students during quizzes and/or exams.
3. Specific Parameters: Discussion on Piazza is strongly encouraged, outside of quizzes and exams. Just be careful about posting any code snippets, try to avoid doing this, and if you can't avoid it, you may be better off emailing the instructor.

LEARNING ACCOMMODATIONS

Stevens Institute of Technology is dedicated to providing appropriate accommodations to students with documented disabilities. The Office of Disability Services (ODS) works with undergraduate and graduate students with learning disabilities, attention deficit-hyperactivity disorders, physical disabilities, sensory impairments, psychiatric disorders, and other such disabilities in order to help students achieve their academic and personal potential. They facilitate equal access to the educational programs and opportunities offered at Stevens and coordinate reasonable accommodations for eligible

students. These services are designed to encourage independence and self-advocacy with support from the ODS staff. The ODS staff will facilitate the provision of accommodations on a case-by-case basis.

For more information about Disability Services and the process to receive accommodations, visit https://www.stevens.edu/office-disability-services. If you have any questions please contact: Phillip Gehman, the Director of Disability Services Coordinator at Stevens Institute of Technology at pgehman@stevens.edu or by phone 201-216-3748.

Disability Services Confidentiality Policy

Student Disability Files are kept separate from academic files and are stored in a secure location within the Office of Disability Services. The Family Educational Rights Privacy Act (FERPA, 20 U.S.C. 1232g; 34CFR, Part 99) regulates disclosure of disability documentation and records maintained by Stevens Disability Services. According to this act, prior written consent by the student is required before our Disability Services office may release disability documentation or records to anyone. An exception is made in unusual circumstances, such as the case of health and safety emergencies.

INCLUSIVITY

Name and Pronoun Usage

As this course includes group work and class discussion, it is vitally important for us to create an educational environment of inclusion and mutual respect. This includes the ability for all students to have their chosen gender pronoun(s) and chosen name affirmed. If the class roster does not align with your name and/or pronouns, please inform the instructor of the necessary changes.

Inclusion Statement

Stevens Institute of Technology believes that diversity and inclusiveness are essential to excellence in academic discourse and innovation. In this class, the perspective of people of all races, ethnicities, gender expressions and gender identities, religions, sexual orientations, disabilities, socioeconomic backgrounds, and nationalities will be respected and viewed as a resource and benefit throughout the semester. Suggestions to further diversify class materials and assignments are encouraged. If any course meetings conflict with your religious events, please do not hesitate to reach out to your instructor to make alternative arrangements.

You are expected to treat your instructor and all other participants in the course with courtesy and respect. Disrespectful conduct and harassing statements will not be tolerated and may result in disciplinary actions.

MENTAL HEALTH RESOURCES

Part of being successful in the classroom involves a focus on your whole self, including your mental health. While you are at Stevens, there are many resources to promote and support mental health. The Office of Counseling and Psychological Services (CAPS) offers free and confidential services to all enrolled students who are struggling to cope with personal issues (e.g., difficulty adjusting to college or trouble managing stress) or psychological difficulties (e.g., anxiety and depression) and who can visit the office in person. CAPS is open from 9:00 am – 5:00 pm Mondays, Wednesdays, Thursdays and Fridays and from 9:00 am – 7:00 pm on Tuesdays during the Fall and Spring semesters; appointments are highly encouraged. For those students who cannot visit the Stevens campus for an in-person appointment, you can contact a local mental health care provider for an in-person appointment, or if you are enrolled in the Stevens Student Health Insurance, you may call Care Connect for 24/7 mental health support at 1-888-857-5462.

EMERGENCY INFORMATION

In the event of an urgent or emergent concern about the safety of yourself or someone else in the Stevens community, please immediately call the Stevens Campus Police at 201-216-5105 or on their emergency line at 201-216-3911. These phone lines are staffed 24/7, year round. For students who do not reside near the campus and require emergency support, please contact your local emergency response providers at 911 or via your local police precinct. Other 24/7 national resources for students dealing with mental health crises include the National Suicide Prevention Lifeline (1-800-273-8255) and the Crisis Text Line (text “Home” to 741-741). If you are concerned about the wellbeing of another Stevens student, and the matter is *not* urgent or time sensitive, please email the CARE Team at care@stevens.edu. A member of the CARE Team will respond to your concern as soon as possible.